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Applicability of Global Hemostatic Tools for Evaluation of Hemostatic State and Detection of Thrombosis in Cats With Cardiomyopathies

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ABSTRACT

Background: Arterial thromboembolism (ATE) is a known complication of cardiomyopathies (CM) in cats. Hypercoagulability is believed to be one predisposing factor.

Objective: Assess applicability of global hemostatic tests for evaluating the systemic hemostatic state in cats with CM with concurrent ATE (CM + ATE).

Animals: Six cats with CM + ATE, 11 cats with CM, and 13 healthy cats.

Methods: Prospective case–control study (2016–2020). Echocardiography, whole blood thromboelastography (TEG), and thrombin generation testing (TGT) were performed. Hypercoagulability was defined for TEG as shortened reaction time (R), shortened kinetics (K), increased angle, or increased maximal amplitude and for TGT as decreased lag time, decreased time to peak, increased peak, or increased endogenous thrombin potential (ETP). If ≥ 1 parameter showed significant hypercoagulability compared with healthy cats, and no parameter showed significant hypocoagulability, the patient was considered hypercoagulable. The opposite criteria defined hypocoagulability.

Results: Hypocoagulability was detected for cats with CM + ATE compared with both cats with CM and healthy cats, characterized by significantly prolonged R and K on TEG as well as significantly increased time-to-peak and decreased peak and ETP on TGT. Additionally, some thrombocytopenia was considered likely in cats with CM + ATE, but could not be fully evaluated because of a high prevalence of concurrent platelet aggregation.

Conclusions and Clinical Importance: A systemic hypocoagulable state was detected in cats with CM with concurrent ATE. Similar hemostatic paradoxes in human patients with thrombosis have been suggested to be related to platelet exhaustion, local endocardial hypercoagulability, or alterations in regulatory proteins.

Abbreviations: aPTT, activated partial thromboplastin time; ARVC, arrhythmogenic right ventricular cardiomyopathy; ATE, arterial thromboembolism; BSH, British shorthair; CM, cardiomyopathy; CV, coefficient of variation; DCM, dilated cardiomyopathy; DLH, domestic longhair; DSH, domestic shorthair; ETP, endogenous thrombin potential; HCM, hypertrophic cardiomyopathy; Hct, hematocrit; K, kinetics; LA, left atrium; LAA, left atrial appendage; MA, maximum amplitude; MC, Maine coon; MRTG, maximum rate of thrombin generation; NETS, neutrophil extracellular traps; NFO, Norwegian forest cat; NSCM, non-specific cardiomyopathy; PLT, platelet count; PT, prothrombin time; R, reaction time; RCM, restrictive cardiomyopathy; SAA, serum amyloid A; SFO, Siberian forest cat; TAT, thrombin-antithrombin; TEG, thromboelastography; TF, tissue factor; TFPI, tissue factor pathway inhibitor; TGT, thrombin generation testing; TMRTG, time to MRTG; TTG, total thrombin generation; WB, whole blood; WB-TEG, whole blood thromboelastography.

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1 | Introduction

Cardiomyopathy (CM) is the most common form of heart disease in cats, with hypertrophic cardiomyopathy (HCM) having a prevalence of approximately 15% in adult cats [1, 2]. Less commonly, cats may present with restrictive (RCM), dilated (DCM), arrhythmogenic right ventricular cardiomyopathy (ARVC), or various non-specific (NSCM) cardiomyopathy phenotypes [2, 3]. Although congestive heart failure is the expected ultimate result [4], a notable proportion of cats (approximately 15%) [2, 4, 5] experience an alternative life-threatening complication, arterial thromboembolism (ATE), concurrently with or even preceding fulminant heart failure [6]. Arterial thromboembolism represents failure of the hemostatic system induced by the underlying heart disease. Thrombus formation generally begins in the left atrium (LA) or its appendage (LAA), which explains the arterial localization of resultant emboli. Once dislodged, emboli most commonly lodge in the aortic bifurcation or, less commonly reported, in the vasculature of one hind or front limb or in the vasculature of organs [7, 8]. Acute paralysis is the typical presentation of limb ATE, with ischemic pain, cyanosis, and lack of palpable pulse in the affected extremities [9]. Although some cats may recover spontaneously or with conservative treatment focusing on the underlying cardiac disease and pain management, treatment otherwise aims at either surgical or rheolytic embolectomy [10, 11] or at medical thrombolysis [12, 13] which is known to carry a high risk of complications, including fatal hemorrhage and reperfusion injury. Additionally, up to 75% of cats experience ATE recurrence [14]. Survival to discharge is reported at 37%–45% [7, 9, 15], with many cats euthanized because of severe clinical signs and poor prognosis, whereas median survival times of 94–117 days have been reported for cats surviving the initial critical days [9, 16]. Hence, preventative treatment is highly desirable. A significantly lowered risk of recurrent ATE has been established when treatment with clopidogrel is initiated [14, 17], a platelet inhibitor shown to decrease platelet aggregation [18]. Today, clopidogrel treatment is recommended for any cat with CM deemed at high risk of ATE, potentially in combination with a factor X inhibitor [17, 19]. Nevertheless, ATE remains common, and further insight into its pathogenesis is necessary, as are biomarkers with the ability to predict its risk in individual cats.

Thrombosis is believed to arise from a combination of altered blood flow, endothelial injury or dysfunction, and hypercoagulability [7, 20]. For cats with CM, stagnant blood flow, especially in the LAA, has been associated with an increased risk of spontaneous echocardiographic contrast in cats [21] and is directly associated with thromboembolism in humans [22, 23], with the stasis believed to both impair clearance of activated coagulation factors and enhance interactions between platelets and endothelium [24]. Endothelial injury or dysfunction in cats with CM is believed to arise from atrial stretch and altered shear stress [6, 25, 26] and may contribute to thrombosis through exposure of sub-endothelial tissue factor (TF) and, thereby, result in increased local thrombin generation [27]. However, changes in the circulating blood supporting hypercoagulability have only been indicated in these cats. Increased platelet activation has been shown [28], but hypercoagulability also involves a systemic imbalance in coagulation factors and their inhibitors, favoring pathological thrombin generation [29]. A few studies have investigated coagulation factors and inhibitors in cats with HCM [25, 29], but have focused mainly on parameters evaluating only isolated parts of hemostasis. Convincing hypercoagulability has not been demonstrated. Therefore, in order to achieve a more complete understanding of the hemostatic state of affected cats, global hemostatic testing may be required [30, 31] to provide information about clot formation dynamics, clot strength properties, clot stability and ultimately fibrinolysis in a low shear environment [32, 33]. Characterizing the hemostatic state of cats with ATE might identify new treatment targets as well as a possibility for personalized medicine with global hemostatic assays as tools to monitor treatment efficiency and refine treatment for the individual cat.

Our objective was 1) to assess the applicability of the global hemostatic tests thromboelastography (TEG) and thrombin generation testing (TGT) for the evaluation of the systemic hemostatic state in cats with ATE and 2) to compare global hemostatic test parameters of cats with CM and ATE to those of cats with uncomplicated CM and healthy cats. We hypothesized that cats with ATE would be in a hypercoagulable state, confirmed by TEG and TGT.

2 | Materials and Methods

Our prospective case–control study was performed at the University Hospital for Companion Animals, University of Copenhagen, Denmark. The study protocol was approved by the Ethical and Administrative Committee at the Department of Veterinary Clinical Sciences, University of Copenhagen, Denmark (07-03-2016, reference number 2016-5), and informed client consent was obtained for all enrolled cats. The enrollment period extended from October 2016 to June 2020.

Cats with CM were included as they presented to the cardiology or emergency services at the University Hospital for Companion Animals. They were divided into a group with CM and a group with CM complicated by ATE. The CM group consisted of cats for which CM was confirmed on echocardiography, but for which no history, present clinical signs, or echocardiographic findings indicated ATE. The CM with concurrent ATE (CM + ATE) group consisted of cats with confirmed CM on echocardiography, and for which an episode of thrombosis was strongly suspected based on history, clinical, echocardiographic, and, when available, necropsy findings.

Patients were excluded from either group if they weighed < 3 kg, if unrelated, non-cardiac comorbidities were discovered on clinical examination, CBC, serum biochemistry, or serum thyroxine concentration, if they had known hemostatic disorders, had been supplemented with antithrombotic medications within the past week, or if study participation was deemed a risk to the cat based on its clinical condition.

Client-owned healthy cats > 1 year of age were included as a control group through a HCM screening program conducted by the cardiology service. These cats were deemed healthy based on history, physical examination, CBC, serum biochemistry, and serum thyroxine concentration as well as an unremarkable cardiac examination including ECG and echocardiography.

2.1 | Echocardiography

An echocardiographic examination was performed, and an ECG (50 mm/s, 10 mm/mV) was recorded for all cats. Images were digitally stored for subsequent measurements. Cats were examined in right and left lateral recumbency, or if in respiratory compromise, in a standing position. Echocardiography was performed based on current recommendations [19, 34] using two-dimensional echocardiography, color-flow Doppler, spectral Doppler, M-mode, and tissue Doppler imaging to document the potential presence of CM as well as signs of ATE.

Cardiomyopathies were phenotypically categorized as HCM, RCM, DCM, ARVC, or NSCM, and subsequently staged in accordance with the current American College of Veterinary Internal Medicine consensus guidelines [19]. Hypertrophic cardiomyopathy was defined as measured left ventricular wall thickness > 6 mm on B-mode or M-mode images using the leading edge-to-leading-edge technique [35]. Presence of left-sided congestive heart failure was determined based on a combination of clinical findings, radiographic findings suggestive of pleural effusion or pulmonary edema, and sonographic findings suggestive of left ventricular restrictive filling pattern [36], pleural effusion, pericardial effusion, and presence of B-lines on lung ultrasound evaluation [37–39]. Right-sided congestive heart failure was determined based on a combination of clinical findings, subjective sonographic findings of marked right atrial enlargement, enlargement of the caudal vena cava, and presence of peritoneal effusion.

Arterial thromboembolism was diagnosed by at least one of the following criteria being fulfilled in cats with echocardiographically confirmed advanced cardiomyopathy: (1) clinical signs (for limb ATE: limb paresis or paralysis, cold affected limb or limbs, absence of pulse in the affected limb or limbs, cyanotic paw pads, pain, or recent history [within the previous 2 months] compatible with a thromboembolic event) or (2) evidence of thromboembolism (limb or organ) with or without intracardiac thrombus material (evident on either echocardiography or necropsy). For cats that were euthanized because of ATE, permission for immediate necropsy was sought from the owner to confirm the presence of a thrombus.

2.2 | Hemostatic Testing

For hemostatic evaluation, 1 EDTA tube and 3 citrated tubes (3.2%) were collected by jugular venipuncture using a 21-gauge butterfly needle and handled as follows: Platelet count was measured as a part of the CBC performed at presentation, including manual blood smear evaluation. If a low platelet count was detected, it was subjectively estimated whether blood smear evaluation showed a sufficient number of platelet aggregates to exclude severe thrombocytopenia. Hematocrit also was measured as part of the CBC and included in the study because it has been shown to have a potential effect on the maximal amplitude of the TEG [40, 41]. Serum amyloid A (SAA) concentration was measured at presentation to evaluate each patient's inflammatory state, which also potentially may affect hemostasis [41]. Whole blood TEG (WB-TEG) was performed as described below. Remaining citrate-stabilized blood was centrifuged, and plasma collected and frozen at −80°C for later batched analysis of fibrinogen, prothrombin time (PT), activated partial thromboplastin time (aPTT), and TGT analysis. The maximal time of storage at −80°C from sample collection to batch analyses was 4.5 years. Centrifugation for healthy cats and part of the CM group was set to 4400 × g for 3 min at room temperature. However, for part of the CM and CM + ATE groups, centrifugation was set to 2500 × g for 5 min because of a communication error. Consequences of this discrepancy will be discussed below.

Tissue factor-activated WB-TEG was performed using an assay that previously has been reported as a reproducible method in cats [32], the computerized thromboelastograph TEG5000 Hemostasis Analyzer (Hemoscope Corporation) with recombinant human TF (Dade Innovin, Siemens, Erlangen, Germany) at a final dilution of 1:50.000 for activation. Citrated whole blood was allowed to stand for 30 min before being transferred to a preheated, re-calcified TEG cup (37°C, 20 μL 0.2 M $CaCl_2$). The analysis was performed by laboratory technicians highly experienced in TEG analysis and run for 120 min. The final parameters evaluated included reaction time (R, reflecting initiation of coagulation), kinetics (K, reflecting kinetics of clot development), angle (alpha, another measure of kinetics), and maximum amplitude (MA, a measurement of maximum clot strength), as well as the derived velocity curve parameters maximum rate of thrombin generation (MRTG, corresponding to the maximal velocity of clot growth), time from initiation until MRTG (TMRTG), and total thrombin generation (TTG, the area under the curve, corresponding to total clot strength).

Thrombin generation testing was performed in batch analysis of duplicate measurements using the calibrated automated thrombography assay, for which acceptable repeatability has been found with analysis of feline plasma [31]. The measurement interval was set to 10 s, the temperature to 39°C, and the total measured time to 45 min. Eighty μl of citrated plasma was added to 20 μL of either the activator solution (PPP low, 1 pM TF, Triolab, AS, Denmark) or thrombin calibrator solution. Thrombin generation was activated according to the manufacturer's instructions (Thrombinoscope BV, Maastricht, The Netherlands), by adding 20 μL of HEPES buffer containing $CaCl_2$ and fluorogenic substrate, and read by an automated highly specific plate reader. The parameters evaluated included lag time (delay in minutes before initiation of thrombin generation), peak (the maximal thrombin concentration), time to peak, and the endogenous thrombin potential (ETP, the area under the curve, representing the total amount of thrombin generated). In a previous in-house validation of the assay, coefficients of variation (CVs) for these parameters ranged from 2% to 5% (intra-assay) and from 10% to 31% (inter-assay) with the highest CVs for peak and ETP (Abstract from the European College of Veterinary Internal Medicine Congress 2018: Langhorn R et al. Global measurement of coagulation in plasma from healthy cats - a potential tool for detection of thrombosis).

The definitions applied for evaluating the hemostatic state of cats in the CM and CM + ATE groups are found in Table 1, with each parameter evaluated as follows: If ≥ 1 parameter showed significant hypercoagulability compared with the healthy cats, and no parameter showed significant hypocoagulability, the patient

TABLE 1 | Parameters of hyper- or hypocoagulability for cats with cardiomyopathy with or without arterial thromboembolism as compared to healthy cats.

	Hypercoagulability	Hypocoagulability
WB-TEG	↓ R, K, TMRTG	↑ R, K, TMRTG
	↑ α, MA, MRTG, TTG	↓ α, MA, MRTG, TTG
TGT	↓ lag-time, time-to-peak	↑ lag-time, time-to-peak
	↑ peak, ETP	↓ peak, ETP

Abbreviations: Alpha, Angle; ETP, Endogenous thrombin potential; K, Kinetics; LY30/60, Lysis at 30 and 60 min after MA; MA, Maximal amplitude; MRTG, Maximum rate of thrombin generation; R, Reaction time; TGT, Thrombin generation testing; TMRTG, Time to MRTG; TTG, Total thrombin generation; WB-TEG, whole blood TEG.

TABLE 2 | Demographic data for the 30 included cats.

	CM + ATE	CM	Healthy cats
n	6	11	13
Breeds	2 DSH, 1 DLH, 1 Birman, 1 Burmese, 1 Ragdoll	3 MCO, 3 BSH, 2 NFO 2 DSH, 1 SIB	5 DSH, 4 NFO, 2 MCO, 2 BSH
Age (years)	8.5 (3–14.25)	9 (2–10)	3 (2–4)
Sex	4 males, 2 females	10 males, 1 female	5 males, 8 females

Note: Ages are presented as median (interquartile range).
Abbreviations: ATE, arterial thromboembolism; BSH, British shorthair; CM, cardiomyopathy; DLH, domestic longhair; DSH, domestic shorthair; MCO, Maine coon; NFO, Norwegian forest cat; SIB, Siberian forest cat.

was considered hypercoagulable. If ≥1 parameter showed significant hypocoagulability compared with the healthy cats, and no parameter showed significant hypercoagulability, the patient was considered hypocoagulable.

2.3 | Statistical Methods

Normality of data was tested using the Shapiro–Wilk test. Distributions between groups (e.g., sex) were examined using the Chi-square test. Comparison between groups was performed using unpaired analysis of variance (for parametric data) or Kruskal-Wallis test (for non-parametric data that did not have parametric distribution with log transformation), followed by individual post hoc analyses. Correlations between TEG velocity curves and TGT, both measures of thrombin generation, were evaluated using Spearman's correlation coefficient. Analyses were conducted using GraphPad Prism version 10 (GraphPad Software, San Diego, CA). Statistical significance was defined as $p < 0.05$.

3 | Results

Thirty cats were included in the study. Eleven cats were included in the CM group, six cats in the CM + ATE group, and 13 healthy cats in the control group (see Table 2 for demographic data). The CM + ATE group was significantly older compared with the control group ($p = 0.01$), but not compared with the CM group, and a significant difference was found between the sex distribution of the groups ($p = 0.03$).

Of cats in the CM group, nine were diagnosed with HCM, and two had NSCM. Six of these cats were presented with mild to moderate disease and five with advanced disease. Of cats in the ATE group, three were diagnosed with HCM, one with DCM, one with ARVC, and one with NSCM. Two of these cats presented with saddle thrombi (bilateral hind limb paralysis), one with a femoral thrombus (unilateral hind limb paralysis), one with echocardiography and necropsy findings compatible with cardiac infarction, and two with a history consistent with recovery from recent ATE episodes (one front limb paralysis and one bilateral hind limb paralysis). The four animals that presented with signs of acute ATE were euthanized. Necropsy confirmed the findings in three cats, but the owner of one cat with a saddle thrombus presentation declined necropsy. This cat had DCM stage C with echocardiographic findings of thromboembolic material in the left ventricle. For details regarding echocardiography, ECG, and necropsy findings of individual cats, see Table S1.

A significant difference was found between measured platelet counts of the groups, with the CM + ATE group having a lower measured platelet count compared with cats with CM but not healthy cats (Table 3). All cats with a measured platelet count (PLT) below reference (1/11 cats with CM, 5/6 cats with CM + ATE, and 5/12 healthy cats) had platelet aggregates noted on blood smears (Table S2), with each cat having a subjective evaluation that severe thrombocytopenia could be excluded based on the number and size of platelet aggregates present. Prothrombin time and aPTT were not significantly different between groups. Fibrinogen concentration was significantly higher in the CM + ATE group compared with the control group, whereas no significant difference was found between groups for SAA. No significant difference was found between groups for hematocrit (Table 3).

A hypocoagulable state was detected in cats with CM + ATE compared with both cats with CM and healthy cats, characterized by prolonged R and K on WB-TEG, corresponding to a significantly slower initiation and development of coagulation in cats with CM + ATE (Figure 1, Table 4). These findings were most pronounced in the cats with a measured PLT $\leq 100 \times 10^9$/L, however, the lowest PLT did not correspond to the most hypocoagulable parameters on WB-TEG. Additionally, cats with CM + ATE had a significantly decreased angle on WB-TEG compared with healthy cats, but not cats with CM. The velocity curves and TGT were in agreement with this hypocoagulability, showing an increased TMRTG and time-to-peak and decreased MRTG, peak, and ETP, corresponding to a significantly slower rate and amount of thrombin generation in cats with CM + ATE (Figure 2, Tables 5 and 5). Hypocoagulability also was detected for the CM group

TABLE 3 | Platelet count, coagulation, and acute-phase parameters for cats with cardiomyopathy with and without arterial thromboembolism as well as healthy control cats.

	CM + ATE	CM	Healthy cats	Reference	*p* value
PLT (10^9/L)	97.5 (62.5–221.8)	284 (228–368)	195 (155–256.3)	190–430	0.01* 0.32** 0.01*** 0.10****
No. of cats with measured PLT ≤ 100 (10^9/L)[a]	4/6	0/11	1/12[b]	—	—
No. of cats with PLT aggregates	5/6	1/11	5/12[b]	—	—
Hct (%)	32.5 (27.93–42.25)	36 (34–38)	36.3 (31.75–41.5)	24–45	0.96*
PT (s)	10.5 (9.7–12.0)	10.3 (9.9–10.6)	9.6 (9.3–10.6)	□	0.30*
aPTT (s)	18.3 (15.4–32.2)	18.4 (14.8–37.8)	18.0 (14.9–24.0)	□	0.62*
Fibrinogen (g/L)	4.3 (1.8–11.2)	1.9 (1.7–2.5)	1.7 (1.4–2.0)	0.5–3.0	0.03* 0.03** 0.36*** 0.65****
SAA (mg/L)	23.2 (0–205.9)	0 (0–0.1)	0.1 (0–0.3)	0–5.0	0.18*

Note: Numbers are presented as median (interquartile range). * signifies the combined *p* value, ** the *p* value between the CM + ATE group and healthy cats, *** the *p* value between the CM + ATE and CM groups, and **** the *p* value between the CM group and healthy cats. □ refers to comparison to simultaneous analysis of feline plasma pool.
Abbreviations: aPTT, activated partial thromboplastin time; ATE, arterial thromboembolism; CM, cardiomyopathy; Hct, hematocrit; PLT, platelet count; PT, prothrombin time; SAA, serum amyloid A.
[a]All cats with measured PLT ≤ 100 × 10^9/L also had PLT aggregates.
[b]For 1/13 healthy cats, the EDTA tube had clotted, and hence a CBC was not available for data analysis. For the remaining cats, every cat in each group with a PLT below reference had sufficient platelet aggregates confirmed on cytology for severe thrombocytopenia to be excluded.

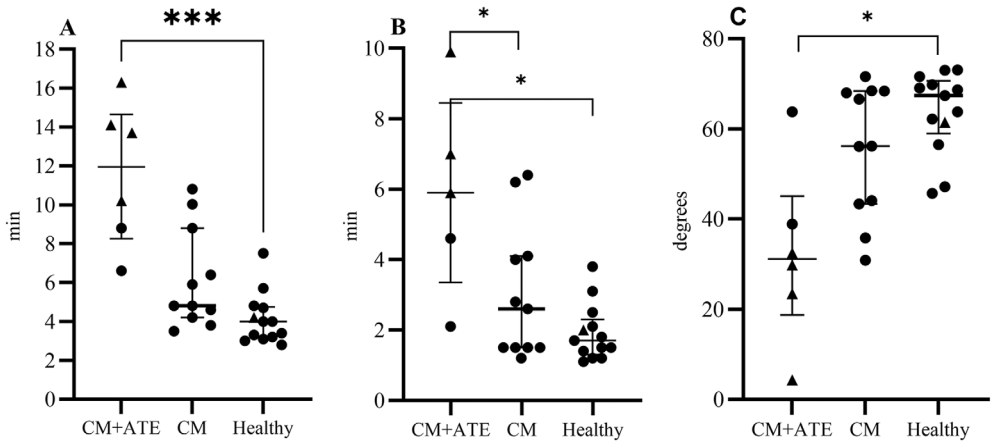

FIGURE 1 | Whole blood thromboelastography parameters reaction time (R) (A), kinetics (K) (B) and angle (α) (C) for cats with cardiomyopathy (CM) with and without arterial thromboembolism (ATE) as well as healthy cats. Medians are shown as horizontal lines. Significant differences ($p < 0.05$) between groups are symbolized with * (between two groups) or *** (between all groups).

compared with the healthy cats, characterized by a prolonged R. No significant differences were found between the remaining TEG and TGT parameters for the CM group and healthy cats (Figures 1 and 2, Tables 4 and 5).

Correlation analyses for WB-TEG velocity curves and TGT identified a significant positive correlation between MRTG and peak ($r = 0.64$, $p < 0.001$) and between TMRTG and time-to-peak ($r = 0.73$, $p < 0.001$), but not between TTG and ETP ($r = -0.2$, $p = 0.29$).

4 | Discussion

We evaluated the systemic hemostatic state of cats with CM and ATE using global hemostatic tests. The original hypothesis

TABLE 4 | Whole blood thromboelastography results (including velocity curves) for cats with cardiomyopathy with and without arterial thromboembolism as well as healthy control cats.

	CM + ATE	CM	Healthy cats	*p* value
R (min)	12.0 (8.3–14.7)	4.8 (4.4–8.8)	4.0 (3.2–4.8)	< 0.001* < 0.001** 0.002*** 0.04****
K (min)	5.9 (3.4–8.5)	2.6 (1.5–4.1)	1.7 (1.3–2.3)	0.002* 0.001** 0.04*** 0.23****
α (°)	31.2 (18.7–45.1)	56.2 (43.4–68.4)	67.4 (59–70.7)	0.004* 0.003** 0.13*** 0.40****
MA (mm)	57.0 (35.8–75.5)	45.1 (40.3–59.5)	54.6 (42.4–58.0)	0.63*
LY30 (%)	0 (0–3.8)	3.1 (0.1–22.0)	3.5 (0.8–14.9)	0.09*
LY60 (%)	0.1 (0–11.0)	10.3 (2.9–31.5)	10.1 (5.3–28.2)	0.06*
MRTG (mm/min)	3.5 (1.9–6.4)	8.0 (5.6–13.8)	12.1 (8.4–15.5)	0.01* 0.004** 0.06*** 0.40****
TMRTG (min)	15.3 (11.4–21.2)	7.3 (5.3–12.0)	5.3 (4.2–6.3)	< 0.001* < 0.001** 0.002*** 0.08****
TTG (mm/min)	706 (448–915)	538 (476–718)	626 (504–690)	0.38*

Note: Numbers are presented as median (interquartile range). * signifies the combined *p* value, ** the *p* value between the CM + ATE group and healthy cats, *** the *p* value between the CM + ATE and CM groups, and **** the *p* value between the CM group and healthy cats.
Abbreviations: Alpha, angle; ATE, arterial thromboembolism; CM, cardiomyopathy; K, kinetics; LY30/60, lysis at 30 and 60 min after MA; MA, maximal amplitude; MRTG, maximum rate of thrombin generation; R, reaction time; TMRTG, time to MRTG; TTG, total thrombin generation.

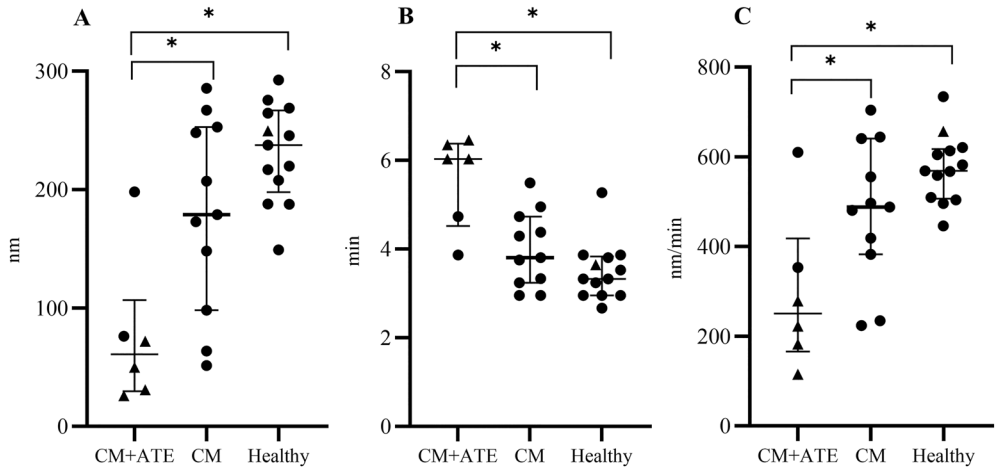

FIGURE 2 | Thrombin generation testing parameters peak (A), time to peak (B), and endogenous thrombin potential (ETP) (C) for cats with cardiomyopathy (CM) with and without arterial thromboembolism (ATE) as well as healthy cats. Medians are shown as horizontal lines. Significant differences ($p < 0.05$) between groups are symbolized with *.

of cats being in a hypercoagulable state was based on a common pathophysiological understanding of thrombosis as well as on studies indicating hypercoagulability in cats with HCM and ATE based on various single parameters of hemostasis [25, 29]. However, both WB-TEG and TGT indicated instead a systemic hypocoagulable state in cats with CM + ATE, as well

TABLE 5 | Thrombin generation test parameters for cats with cardiomyopathy with and without arterial thromboembolism as well as healthy control cats.

	CM + ATE	CM	Healthy cats	*p* value
Lag time (min)	2.7 (2.0–3.4)	2.3 (1.8–2.7)	1.8 (1.6–2.1)	0.06*
Peak (nm)	61.0 (29.7–106.7)	179 (98.2–252.8)	238.0 (197.7–266.7)	< 0.001* < 0.001** 0.002*** 0.18 ****
Time to peak (min)	6.0 (4.5–6.4)	3.8 (3.2–4.7)	3.3 (3.0–3.8)	< 0.001* < 0.001** 0.01*** 0.26****
ETP (nm/min)	251 (166–418)	488 (383–641)	569 (507–617)	< 0.001* < 0.001** 0.03*** 0.20****

Note: Numbers are presented as median (interquartile range). * signifies the combined *p* value, ** the *p* value between the CM + ATE group and healthy cats, *** the *p* value between the CM + ATE and CM groups, and **** the *p* value between the CM group and healthy cats.
Abbreviations: CM, cardiomyopathy; CM + ATE, cardiomyopathy with arterial thromboembolism; ETP, endogenous thrombin potential.

as similar but less pronounced findings in cats with CM alone. Interestingly, our findings are in agreement with a recent study applying automated viscoelastic coagulation monitoring in cats with ATE, which found a subset of cats to have hypocoagulable tracings [42].

This paradoxical hemostatic finding does not change the fact that these cats present with thrombosis and should be treated accordingly. Current consensus guidelines recommend clopidogrel treatment in any cat deemed at high risk of ATE [19], based on a confirmed lowered risk of recurrent ATE with this treatment [14, 17]. Nevertheless, our findings suggest reevaluation of systemic hypercoagulability as a necessary triggering cause of thrombosis in cats with CM and illustrate that hemostasis is a dynamic, complex, and balanced process with prothrombotic, fibrinolytic, and regulatory mechanisms constantly in play.

Cats with CM + ATE had significantly higher fibrinogen concentrations than healthy cats, mirroring previous research [29]. In fact, fibrinogen is one of the parameters previously suggested to reflect a hypercoagulable state in these cats [29]. However, fibrinogen is also a positive acute phase protein and increases primarily in inflammatory states. For SAA, another acute phase protein, no significant difference was found between groups, but 50% of the CM + ATE group had increased SAA concentrations compared with only 1 cat with CM and no healthy cats, and the lack of significant difference was likely a consequence of sample size. Inflammation and hemostasis are strongly linked [43], and the contribution of proinflammatory mechanisms to the development of thrombosis in these cats is possible. Increased circulating markers of neutrophil extracellular traps (NETS) have been found in cats with CM and ATE, and NETS have been found histopathologically in the emboli themselves [44, 45]. The formation of NETS in ATE is suggested to be a consequence of endothelial injury and platelet activation, and NETS themselves are known to be thrombogenic with their ability to trap circulating components, including coagulation factors [44, 45]. Nevertheless, inflammation also may arise as a result of the thrombotic state, induced by the severe tissue ischemia known to occur with ATE [46], making causation challenging to establish. Notably, inflammation is known to induce systemic hypercoagulability [41], not hypocoagulability, and therefore, inflammation as measured by acute phase proteins alone does not explain our results.

Other parameters previously suggested to reflect hypercoagulability in cats with ATE are thrombin-antithrombin (TAT) complexes and D-dimers [25, 29]. Thrombin-antithrombin complexes represent net activation of coagulation because antithrombin rapidly binds newly formed thrombin, and an increased amount therefore could signify true hypercoagulability. However, although 25% of cats with HCM had increased TAT in both studies, no significant difference could be found between groups to confirm hypercoagulability as the general hemostatic state of the study populations of HCM cats [25, 29]. On the other hand, D-dimers do not directly reflect hypercoagulability but simply the breakdown of previously formed fibrin. Although this finding naturally can lead to suspicion of an underlying hypercoagulable state, a thrombus may, in some situations, be found in the presence of systemic hypocoagulability [47, 48], as also found in our study. Additionally, the usefulness of D-dimers in cats has been questioned [49, 50].

Generally, when evaluating global hemostatic tests, clot strength is considered the primary parameter of hyper- or hypocoagulability [40]. In our study, this component of coagulation was the least affected, with no significant difference found between groups for MA or TTG on WB-TEG, although the ETP on TGT showed significant differences between cats with CM + ATE and both other groups. Although TTG and ETP reflect only thrombin, MA is known to represent, in part, the patient's fibrinogen concentration [51]. Fibrinogen concentration was increased in cats with CM + ATE and therefore could explain why hypocoagulability was not reflected by this parameter, although ETP did identify weaker clot strength. The MA also is affected by the

PLT and hematocrit [40, 41, 51]. The hematocrit was not significantly different between groups. Multiple cats had a measured low PLT, however. Nevertheless, blood smear evaluation identified platelet aggregates in all of these cats, the numbers of which were considered sufficient to rule out severe thrombocytopenia (Table S2). The high occurrence of low measured PLT and platelet aggregates in the CM + ATE group was an interesting finding. Platelet activation is known to occur in cardiomyopathy, but the same results were not found in the CM group without ATE. A more advanced state of disease, along with stress activation related to the severe pain and discomfort of ATE, might have contributed, as may inflammation. In vitro activation of platelets cannot be excluded, however, and it should be considered a possibility that micro-clots in the blood may have produced the low measured PLT. This possibility could have affected other hemostatic analyses as well. However, all blood collection was performed according to protocol, and all tubes were evaluated before analysis, indicating no signs of in vitro coagulation in the cats with CM + ATE, and it would be unlikely for the same pre-analytical error to have affected 5/6 of the cats specifically in this group. Also, the cats with CM and no ATE, for which PLT were unremarkable, also were in a mild hypocoagulable state, ruling out thrombocytopenia as the only cause of hypocoagulability in our study. Accordingly, it may be more likely that circulating micro-thrombi are a part of the pathogenesis of ATE, leading to some platelet consumption and a high occurrence of aggregation. Interestingly, a study in humans with myocardial infarction identified hyper-reactive platelets in the initial phase, followed by decreasing reactivity over time [52], a phenomenon termed platelet exhaustion. Although this phenomenon has not been identified in cats, it supports that hyper-reactive platelets are likely present in cats and is a plausible explanatory factor for initial thrombosis followed by hypocoagulability. In our study, the four cats with measured PLT $\leq 100 \times 10^9$/L had the most pronounced hypocoagulable tracings, in accordance with a recent study [42]. Nevertheless, the TEG MA was not decreased in this group, supporting the interpretation of platelet aggregates to rule out severe thrombocytopenia.

Despite the possible influence of PLT and altered thrombocyte activity, our findings show that it is not the only factor involved in the hypocoagulability detected. In fact, the parameters mainly affected were those primarily reflecting coagulation factors (R and K on the WB-TEG as well as its velocity curve and the plasma-based TGT parameters). Of note, these may be affected by platelets because platelets provide the procoagulant surface for thrombin generation, but only a mild influence on the parameters has been found by modulating platelet function [53]. The coagulation factors themselves are of highest importance, which could suggest coagulation factor consumption during the ATE episode with subsequent hypocoagulability. This mechanism would reflect the pathophysiology of disseminated intravascular coagulopathy [54] and also has been shown in a study of dogs with thrombosis, although these dogs did not have a concurrent low ETP [55]. However, neither of these suggested hypotheses would explain why the cats with CM alone showed similar, albeit milder, hypocoagulability at a stage when no signs of thrombosis were yet detected. Either changed platelet activity or coagulation factor consumption or both would have to occur in the earlier stages leading up to ATE rather than as a result of the thrombosis itself.

Hence, another potential cause could be a difference between what is measurable systemically and what happens locally. It could be that these cats have a local endocardial hypercoagulability and a concurrent or resultant systemic hypocoagulability [56]. Local hypercoagulability most likely would be a result of endothelial injury, believed to arise from atrial stretch and altered shear stress. Endothelial injury has been confirmed in the endocardium in cats with HCM at necropsy [57], and one study showed an increased concentration of plasma von Willebrand factor antigen in cats with HCM, suggesting this finding as evidence of release from injured endothelial cells [29]. It has been shown experimentally in rabbits, mice, and rats that local hypercoagulability exists, identified as extracellular TF expression on endothelial cells and decreased expression of endothelial thrombomodulin [26, 58, 59]. Thrombomodulin is an endothelial luminal glycoprotein that functions as a natural anticoagulant. Lack of thrombomodulin in the endocardium leads to decreased activation of protein C and increased local thrombin production, in accordance with the hypothesis of local hypercoagulability [26]. However, significantly lower circulating thrombomodulin concentrations have been identified in cats with CM and enlarged LA [60], which would be more suggestive of systemic rather than local hypercoagulability. Nevertheless, it could potentially result in a consumptive hypocoagulability as the LA progressively enlarges. A regulatory protein potentially involved is tissue factor pathway inhibitor (TFPI). This protein is also a natural anticoagulant, synthesized by vascular endothelium and smooth muscle cells, and serves as the primary inhibitor of TF in hemostasis [58]. Although lack of TFPI will lead to uncontrolled coagulation, paradoxically, it has been shown that TFPI is commonly upregulated in thrombotic states [58]. This effect is believed to be a result of endothelial injury or compensatory upregulation secondary to the prothrombotic state. Also suggesting potential local hypercoagulability, an experimental study in rabbits showed that TFPI concentrations in blood proximal to the site of thrombosis were higher than those collected distally, in accordance with local consumption [61]. We did not measure regulatory proteins such as thrombomodulin or TFPI in cats with and without ATE. However, investigation of the role of these proteins in the systemic hypocoagulability reported here would be valuable.

Our study had several limitations. The use of global hemostatic tests to predict the development of thrombosis in cats started as a proof of concept study with no previous research on which to base a sample size calculation. The study sample size was relatively small, and the results should be confirmed in larger studies. However, a small sample size would be more likely to lead to a risk of type II error with no identifiable significant differences found between groups. Because differences between groups were significant in our study, and TEG and TGT were in agreement, the results are considered reliable. Secondly, the PLT of the cats was difficult to evaluate with certainty because many cats had a low measured PLT and concurrent platelet aggregates. Although evidence of a pre-analytical error could not be detected, in vitro activation and micro-clotting of samples cannot be excluded as a cause of the low measured PLTs. If micro-clotting did occur, it might have affected other hemostatic testing as well. However, the cats with CM and no ATE, for which PLTs were unremarkable, also showed a mild hypocoagulable state. Thirdly, significant differences in age and sex

were found between the study group and healthy controls, with older age and more females among the cats with CM. However, although older age and female sex have been shown to affect coagulation in some species [62–64], no studies have confirmed this possibility in cats, and both of these scenarios would have been likely to contribute to hypercoagulability rather than the hypocoagulability identified in our study. Fourthly, an unfortunate discrepancy was found in how samples had been handled in earlier versus later years of the study. However, because WB-TEG results (which were analyzed at the time of admission) were in agreement with the results of plasma TGT (analyzed at study conclusion), the results are considered reliable. Finally, the initial samples collected in 2016 were subjected to considerably longer storage time than samples collected in 2020, which could have affected the results. However, as stated above, WB-TEG and TGT results were in agreement, just as TGT for cats included earlier and later showed similar patterns. These findings make storage time less likely to have affected the results.

In conclusion, a systemic hypocoagulable state was detected in cats with ATE as opposed to the expected hypercoagulability. A milder hypocoagulable state was found in cats with CM without ATE, suggesting dynamic alterations in the hemostatic capability over time as CM progresses. Although the cause is unclear, platelet exhaustion and local hypercoagulability in the endocardium with subsequent systemic consumption of coagulation factors are among speculated causes. Also, some degree of thrombocytopenia appears to occur in these cats, likely as a result of consumption. Clinically, these findings raise the question of whether normalization of global hemostatic testing would be beneficial for cats with CM and a concurrent hypocoagulable state, and whether normalization is, in fact, paradoxically achieved with antithrombotic treatment in these patients, given that such therapy has been shown to decrease recurrence of ATE. Therefore, additional studies are needed to evaluate the effect of such treatment on global hemostatic tests in cats with CM and whether these tests may serve as monitoring tools, optimizing treatment for each cat.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

Author Lise N. Nielsen is the head of the Veterinary Diagnostic Laboratory at the Department of Veterinary Clinical Sciences, University of Copenhagen. This laboratory is available to veterinary practices throughout Denmark and offers thromboelastography analyses.

Author Jørgen Koch is the head of the cardiology department at the Department of Veterinary Clinical Sciences, University of Copenhagen. This service runs a hypertrophic cardiomyopathy screening program for cats.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Descriptive cardiac data for the seventeen cats with cardiomyopathy with and without arterial thromboembolism. **Table S2:** Measured platelet count (PLT) and descriptive data of platelets on blood smear evaluation for cats with cardiomyopathy (CM), cats with CM and arterial thromboembolism (ATE) and healthy cats.